How to store and route your (physical) mail like a pro

personal edition

Bart Dorlandt

The usual view after vacation

Bart Dorlandt

- Freelance Network Automation Solution Architect
- 20 years in network engineering and automation
- Last year I presented "Repos are like children - Parenting 101"
- Lives by the phrase: *There must be a better way*

Pragmatic approach to a better mail handling

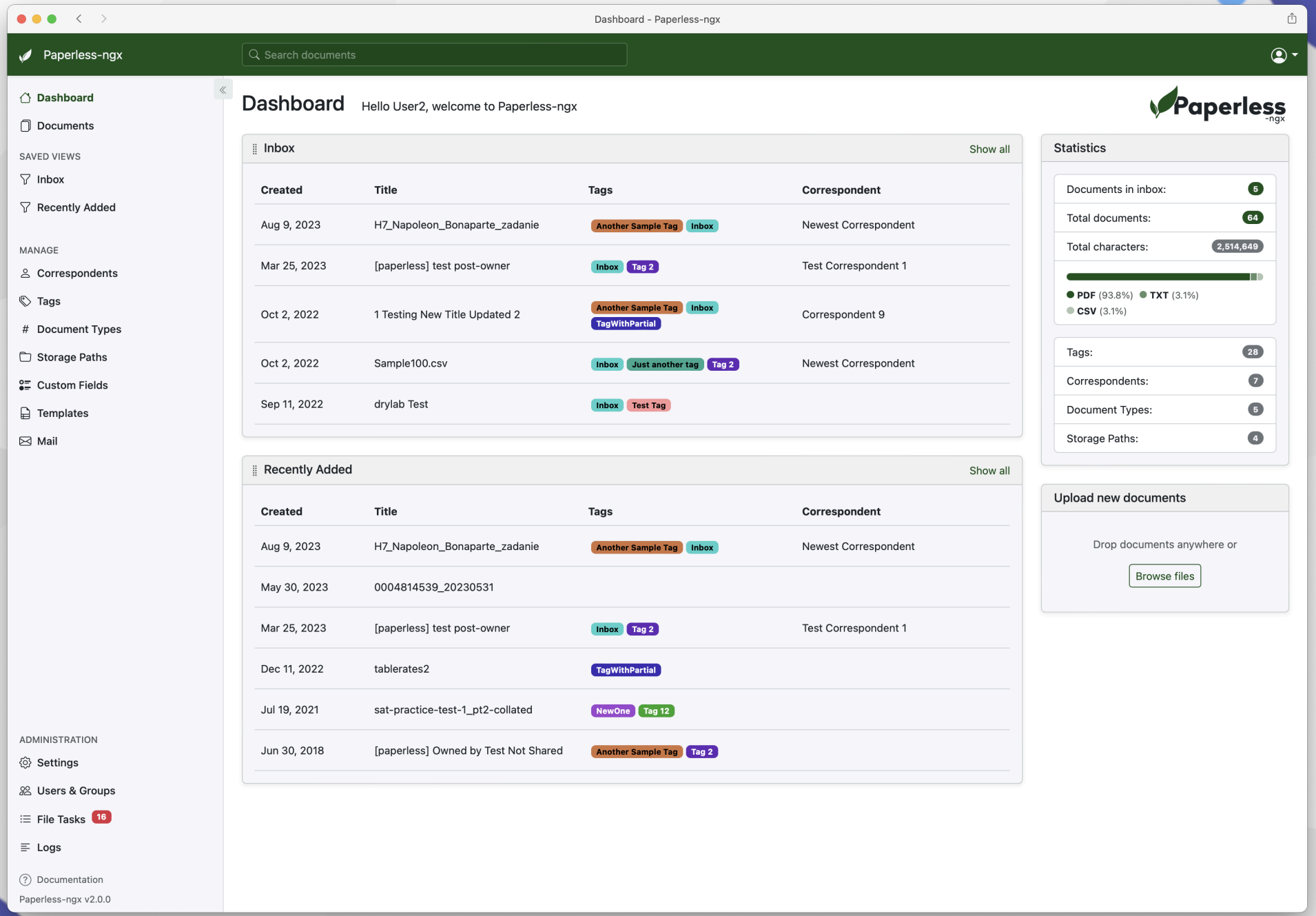
- The problem: physical mail is a mess
- The goal: have a better way to handle it, store it, and find it
- The solution: a mail handling system that is flexible, automated, and easy to use

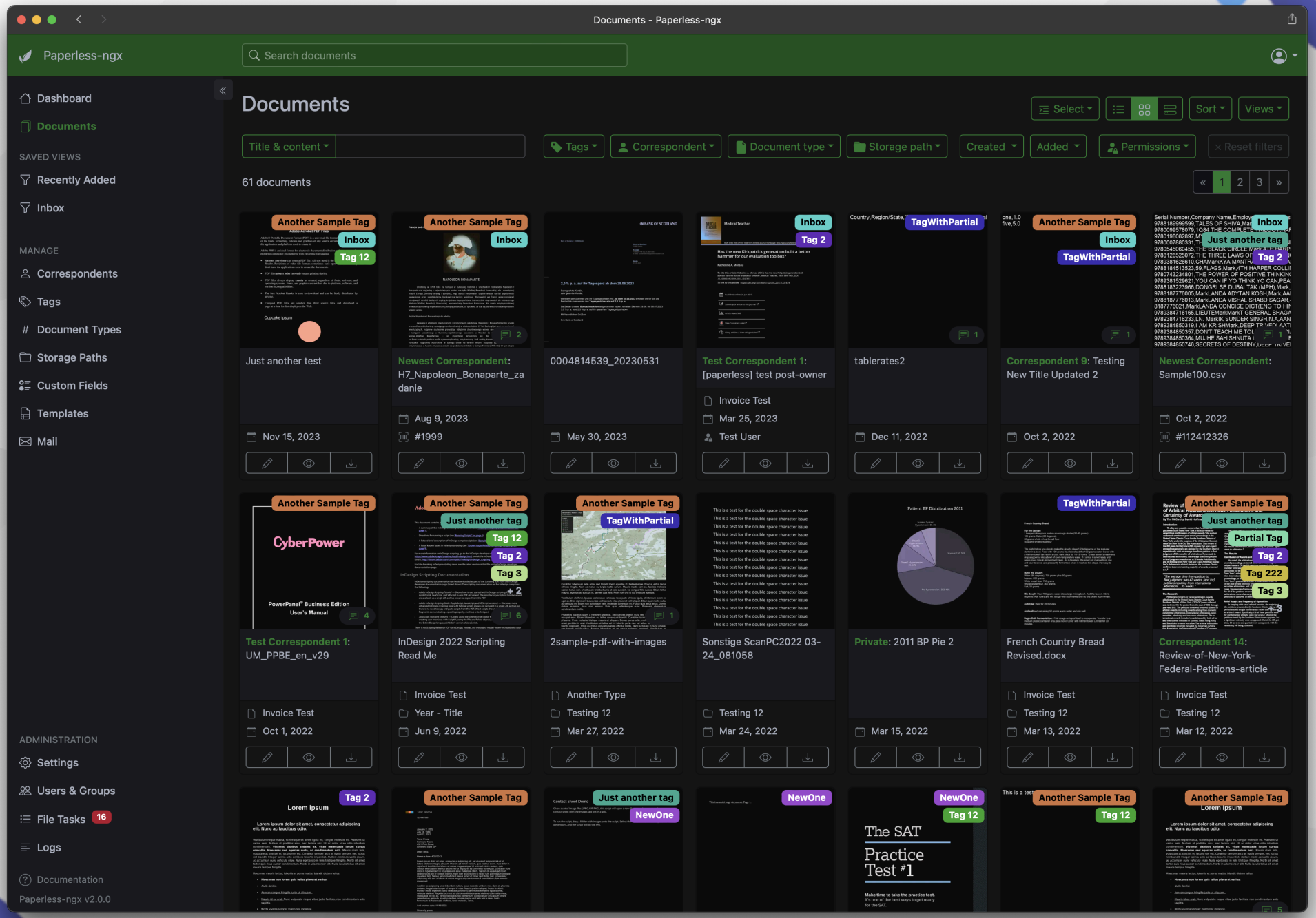
Paperless-ngx

- I had heard about paperless-ngx, an open-source document management system
- I have a few friends that were already happy with it

What is paperless-ngx

- A document management system that allows you to store, organize, and search your documents
- It has a web interface, an API, and can read and parse emails
- It can also watch folders for new documents to process
- It has features for structuring, self-improving & workflows


Dashboard - Paperless-ngx
Paperless-ngx
Search documents
Dashboard
Documents
SAVED VIEWS
Inbox
Recently Added
MANAGE
Correspondents
Tags
Document Types
Storage Paths
Custom Fields
Templates
Mail
ADMINISTRATION
Settings
Users & Groups
File Tasks 16
Logs
Documentation
Paperless-ngx v2.0.0
Dashboard
Hello User2, welcome to Paperless-ngx
Paperless -ngx
Inbox
Show all
Created Title Tags Correspondent
Aug 9, 2023 H7_Napoleon_Bonaparte_zadanie Another Sample Tag Inbox Newest Correspondent
Mar 25, 2023 [paperless] test post-owner Inbox Tag 2 Test Correspondent 1
Oct 2, 2022 1 Testing New Title Updated 2 Another Sample Tag Inbox TagWithPartial Correspondent 9
Oct 2, 2022 Sample100.csv Inbox Just another tag Tag 2 Newest Correspondent
Sep 11, 2022 drylab Test Inbox Test Tag
Recently Added
Show all
Created Title Tags Correspondent
Aug 9, 2023 H7_Napoleon_Bonaparte_zadanie Another Sample Tag Inbox Newest Correspondent
May 30, 2023 0004814539_20230531
Mar 25, 2023 [paperless] test post-owner Inbox Tag 2 Test Correspondent 1
Dec 11, 2022 tablerates2 TagWithPartial
Jul 19, 2021 sat-practice-test-1_pt2-collated NewOne Tag 12
Jun 30, 2018 [paperless] Owned by Test Not Shared Another Sample Tag Tag 2
Statistics
Documents in inbox: 5
Total documents: 64
Total characters: 2,514,649
PDF (93.8%) TXT (3.1%)
CSV (3.1%)
Tags: 28
Correspondents: 7
Document Types: 5
Storage Paths: 4
Upload new documents
Drop documents anywhere or
Browse files


Documents - Paperless-ngx
Paperless-ngx
Search documents
Dashboard
Documents
SAVED VIEWS
Recently Added
Inbox
MANAGE
Correspondents
Tags
Document Types
Storage Paths
Custom Fields
Templates
Mail
ADMINISTRATION
Settings
Users & Groups
File Tasks 16
Logs
Documentation
Paperless-ngx v2.0.0
Documents
Select
Sort
Views
Title & content
Tags
Correspondent
Document type
Storage path
Created
Added
Permissions
Reset filters
61 documents
« 1 2 3 »
Another Sample Tag
Inbox
Tag 12
Just another test
Nov 15, 2023
Newest Correspondent: H7_Napoleon_Bonaparte_zadanie
Aug 9, 2023
#1999
0004814539_20230531
May 30, 2023
Tag 2
Test Correspondent 1: [paperless] test post-owner
Invoice Test
Mar 25, 2023
Test User
TagWithPartial
tablerates2
Dec 11, 2022
Correspondent 9: Testing New Title Updated 2
Oct 2, 2022
Just another tag
Newest Correspondent: Sample100.csv
Oct 2, 2022
#112412326
CyberPower
PowerPanel® Business Edition User's Manual
Test Correspondent 1: UM_PPBE_en_v29
Invoice Test
Oct 1, 2022
InDesign 2022 Scripting Read Me
Year - Title
Jun 9, 2022
Tag 3
2sample-pdf-with-images
Another Type
Testing 12
Mar 27, 2022
Sonstige ScanPC2022 03-24_081058
Mar 24, 2022
Private: 2011 BP Pie 2
Mar 15, 2022
French Country Bread Revised.docx
Mar 13, 2022
Partial Tag
Tag 222
Correspondent 14: Review-of-New-York-Federal-Petitions-article
Mar 12, 2022
Lorem ipsum
NewOne
The SAT Practice Test #1

2sample-pdf-with-images - Paperless-ngx

Paperless-ngx

Search documents

Dashboard
Documents

SAVED VIEWS
Inbox
Recently Added

OPEN DOCUMENTS
2sample-pdf-with-images
H7_Napoleon_Bonaparte_zadanie
Close all

MANAGE
Correspondents
Tags
Document Types
Storage Paths
Custom Fields
Templates
Mail

ADMINISTRATION
Settings
Users & Groups
File Tasks 16
Logs

Documentation
Paperless-ngx v2.0.0

2sample-pdf-with-images

Page 1 of 10 | Delete | Download | Actions | Custom Fields | Share Links

Discard | Save & next | Save

Details | Content | Metadata | Notes 1 | Permissions

Title: 2sample-pdf-with-images

Archive serial number: +1

Date created: 03/27/2022

Correspondent:

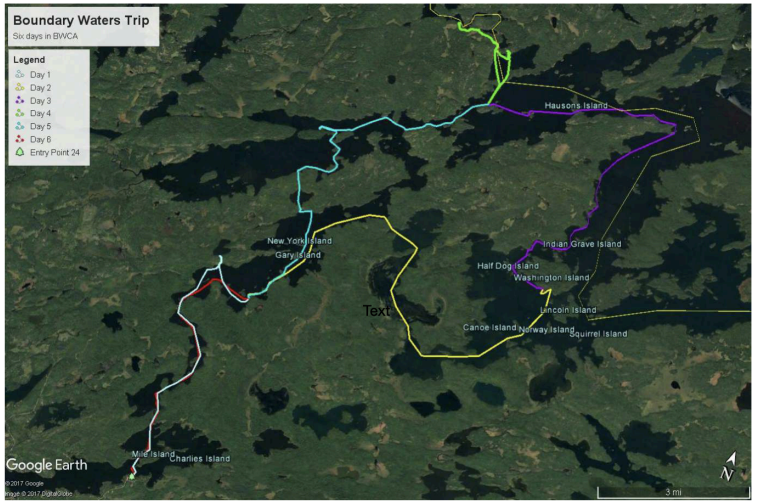
Document type: Another Type

Storage path: Testing 12

Tags: Another Sample Tag

Suggestions: TagWithPartial

Discard | Save & next | Save



Curabitur bibendum ante urna, sed blandit libero egestas id. Pellentesque rhoncus elit in lacus ultrices fringilla. Nam ac metus eu turpis mattis rutrum. Mauris mattis sem ex, facilisis molestie sapien luctus non. Vestibulum tincidunt urna at odio suscipit, vel congue felis cursus. Etiam tellus magna, egestas ac suscipit in, laoreet quis felis. Proin non orci id dui tincidunt egestas.

Vestibulum eleifend, ligula a scelerisque vehicula, risus justo ultricies ligula, et interdum lorem ex eget ex. Duis dignissim lacus vitae velit laoreet, vitae placerat velit aliquet. Etiam eget mollis nulla, ac vehicula mi. Etiam non sollicitudin velit, imperdiet commodo mi. Fusce quis tellus tellus. Donec dictum euismod risus non tempus. Duis quis pellentesque nunc. Praesent elementum condimentum mollis.

Phasellus dapibus quam a hendrerit placerat. Sed ultrices blandit nulla sed sodales. Nunc quis volutpat eros. Etiam bibendum eu tellus consequat blandit. Curabitur lacinia cursus diam sed pharetra. Proin molestie tristique mauris ut aliquam. Donec purus odio, molestie id suscipit sit amet, porttitor in erat. Vestibulum ut tellus vel mi lobortis porta nec vel tellus. Quisque pretium blandit dignissim. Proin eu metus convallis sapien efficitur mollis. Nunc luctus ex in nunc ornare, nec blandit orci faucibus. Aenean bibendum mi vel neque euismod hendrerit. Vestibulum ac pharetra magna. Ut rutrum, orci at blandit faucibus, justo mauris iaculis mauris, ut tempor lectus risus at ligula.

Duis non tincidunt purus. Nam quis sapien risus. Donec mattis convallis tempor. Fusce aliquam aliquet eros, nec rutrum lectus pretium at. Praesent blandit justo a mi dignissim placerat. Ut ullamcorper elit eget diam maximus iaculis. In bibendum in massa eget facilisis. In iaculis lectus

Paperless-ngx

```yaml
services:
  broker:
    image: docker.io/library/redis:7
    restart: unless-stopped
    volumes:
      - redisdata:/data

  webserver:
    image: ghcr.io/paperless-ngx/paperless-ngx:latest
    restart: unless-stopped
    depends_on:
      - db
      - broker
      - gotenberg
      - tika
    ports:
      - "8078:8000"
    volumes:
      - data:/usr/src/paperless/data
      - /.../media:/usr/src/paperless/media
      - /.../export:/usr/src/paperless/export
      - /.../consume:/usr/src/paperless/consume
    env_file: docker-compose.env
    environment:
      PAPERLESS_REDIS: redis://broker:6379
      PAPERLESS_DBHOST: db
      PAPERLESS_TIKA_ENABLED: 1
      PAPERLESS_TIKA_GOTENBERG_ENDPOINT: http://gotenberg:3000
      PAPERLESS_TIKA_ENDPOINT: http://tika:9998
```

```yaml
  db:
    image: docker.io/library/postgres:16
    restart: unless-stopped
    volumes:
      - pgdata:/var/lib/postgresql/data
    environment:
      POSTGRES_DB: paperless
      POSTGRES_USER: paperless
      POSTGRES_PASSWORD: "${POSTGRES_PASSWORD}"

  gotenberg:
    image: docker.io/gotenberg/gotenberg:8.7
    restart: unless-stopped

    # The gotenberg chromium route is used to convert .eml files. We do not
    # want to allow external content like tracking pixels or even javascript.
    command:
      - "gotenberg"
      - "--chromium-disable-javascript=true"
      - "--chromium-allow-list=file:///tmp/.*"

  tika:
    image: docker.io/apache/tika:latest
    restart: unless-stopped

volumes:
  data:
  pgdata:
  redisdata:
```

Destinations

Where do my documents need to go?

- Paperless
- Bookkeeping system (email / app / web)
- Bookkeeper (email)

Input

- I don't want to use this
- I "can't" use the camera app on the phone
  - doesn't work well in the dark
  - doesn't generate a pdf
- It should be easy!

Dropbox camera

- It takes the picture, and:
  - enhances the contrast
  - finds the corners
  - generates a pdf
  - uploads it to the dropbox cloud in a chosen folder

"NOTE: you'll have to accept this is synced to the cloud."

Fahrzeughaus Laufer
Kesselstr. 1
79843 Loeffingen
Freie Tankstelle
07654 / 7137

- StNr. Station: 13185-10202 -

Bon-Nr.: 521204 Datum: 06.05.2026
Verk. : 1 Zeit: 09:52:35

Menge	Einzelpreis	Gesamtpreis
* Diesel 1		Säule 1 ST:D *
* 38,06 Liter	2,019 EUR	76,84 EUR*
*	pro Liter	*

GESAMTSUMME: 76,84 EUR

-K-U-N-D-E-N-B-E-L-E-G-

Terminal-ID : 75087016
TA-Nr 212984 BNr 0122

Kartenzahlung
kontaktlos
DEBIT MASTERCARD

EUR 76,84

Solution dependencies

- Synced folder
- Some code to glue it all together
- Execute the code on a regular basis
- Paperless

What do we need?

- CloudSync to sync the dropbox folder to the NAS
- Docker + docker-compose
- Python
- Task Scheduler

Requirements of this little glueware

- Ability to send to multiple destinations
- Ability to process folders for new files
- Ability to move files to done folder
- Ability to notify me in case of errors

Let's dive in

```python
def main() -> None:
    """Main entry point for the script."""
    # Reading the vars here to ensure env dependencies are present and loaded
    paperless_vars = PaperlessVars(
        api_token=env.str("PAPERLESS_API_TOKEN"),
        api_path=env.str("PAPERLESS_API_PATH"),
        api_url=env.str("PAPERLESS_API_URL"),
    )
    bookkeeping_vars = EmailVars(
        to=env.str("BOOKKEEPING_EMAIL", validate=[validate.Length(min=4), validate.Email()]),
    )
    bookkeeper_vars = EmailVars(
        to=env.str("BOOKKEEPER_EMAIL", validate=[validate.Length(min=4), validate.Email()]),
    )
    ...

if __name__ == "__main__":
    MAIN_PATH = Path(env.str("PROCESS_FOLDER"))
    SMTP_VARS = SmtpVars(
        smtp_srv=env.str("SMTP_SRV"),
        smtp_usr=env.str("SMTP_USR", validate=[validate.Length(min=4), validate.Email()]),
        smtp_pwd=env.str("SMTP_PWD"),
        smtp_port=env.int("SMTP_PORT", default=465),
    )
    ERROR_EMAIL = env.str("ERROR_EMAIL", validate=[validate.Length(min=4), validate.Email()])
    FROM_EMAIL = env.str("SMTP_USR", validate=[validate.Length(min=4), validate.Email()])
    main()
```

```python
@dataclass
class SmtpVars:
    """Email (SMTP) configuration."""

    smtp_srv: str
    smtp_usr: str
    smtp_pwd: str
    smtp_port: int = 465

@dataclass
class PaperlessVars:
    """Paperless API configuration."""

    api_token: str
    api_path: str
    api_url: str

@dataclass
class EmailVars:
    """Email configuration."""

    to: str
```

```python
def main() -> None:
    # 2nd part of the main function, after reading the vars
    paperless_processor = PaperlessAPIProcessor(paperless_vars)
    bookkeeping_processor = EmailProcessor(bookkeeping_vars)
    to_person_processor = EmailProcessor(bookkeeper_vars)

    logger.info("Loaded variables, processing directories...")
    process_folder(MAIN_PATH / "to_paperless", processors=[paperless_processor])
    process_folder(MAIN_PATH / "to_bookkeeping", processors=[bookkeeping_processor])
    process_folder(MAIN_PATH / "to_bookkeeping_paperless", processors=[paperless_processor, bookkeeping_processor])
    process_folder(MAIN_PATH / "to_paperless_bookkeeper", processors=[paperless_processor, to_person_processor])
    process_folder(MAIN_PATH / "to_bookkeeper", processors=[to_person_processor])
```

```python
def process_folder(folder: Path, processors: list[FileProcessor]) -> None:
    """Process files in the given folder with the provided processors.
    Files starting with a dot are ignored.
    """
    folder.mkdir(exist_ok=True)
    for file in list(folder.iterdir()):
        if not file.is_file() or file.name.startswith("."):
            continue

        # Track if any processor succeeded
        processed = 0
        for processor in processors:
            # Only move to done if all processors succeed
            result = processor.process(file)
            processed += result
            if not result:
                error_email(subject="File Processing Error", filename=file.name)

        if processed == len(processors):
            move_to_done(file)
```

```python
def move_to_done(filepath: Path) -> None:
    """Move processed file to the done directory."""
    parent = filepath.parent.name
    done_dir = MAIN_PATH / "done" / parent
    done_dir.mkdir(parents=True, exist_ok=True)
    target = done_dir / filepath.name
    filepath.rename(target)
    logger.info(f"Moved {filepath} to {target}")
```

```python
from typing import Protocol

class FileProcessor(Protocol):
    """Protocol for file processors."""

    def process(self, filepath: Path) -> bool:
        """Process the file and return True if successful, False otherwise."""
```

```python
class PaperlessAPIProcessor:
    """Processor for uploading files to the Paperless API."""

    def __init__(self, vars: PaperlessVars) -> None:
        """Initialize with PaperlessVars."""
        self.vars = vars

    def process(self, filepath: Path) -> bool:
        """Process the file and upload it to the Paperless API."""
        url = f"{self.vars.api_url.rstrip('/')}{self.vars.api_path}"
        headers = {"Authorization": f"Token {self.vars.api_token}"}
        try:
            with filepath.open("rb") as f:
                response = requests.post(url, headers=headers, files={"document": f}, timeout=10)
        except requests.ConnectionError as e:
            logger.error(f"Failed to connect to Paperless API: {e}")
            return False
        except urllib3.exceptions.MaxRetryError as e:
            logger.error(f"Max retries exceeded while connecting to Paperless API: {e}")
            return False
        except urllib3.exceptions.NameResolutionError as e:
            logger.error(f"Name resolution error while connecting to Paperless API: {e}")
            return False
        logger.debug(f"Response from Paperless: {response.status_code} {response.text}")
        if response.status_code == 200:
            logger.info(f"Uploaded to Paperless: {filepath}")
            return True
        else:
            logger.error(f"Failed to upload {filepath} to Paperless: {response.status_code} {response.text}")
            return False
```

```python
class EmailProcessor:
    """Processor for sending files via email to bookkeeping."""

    def __init__(self, vars: EmailVars) -> None:
        """Initialize with EmailVars."""
        self.vars = vars

    def process(self, filepath: Path) -> bool:
        """Process the file and send it via email."""
        msg = new_email_message(
            subject=filepath.name,
            smtp_to=self.vars.to,
        )
        msg.set_content(f"Hi,\n\nPlease find attached the file: {filepath.name}.\n\nBest regards.")
        with filepath.open("rb") as f:
            msg.add_attachment(
                f.read(),
                maintype="application",
                subtype="octet-stream",
                filename=filepath.name,
            )
        try:
            send_email(msg=msg)
        except Exception as e:
            logger.error(f"Failed to send {filepath} via email: {e}")
            return False
        else:
            logger.info(f"Sent successfully via email: {filepath}")
            return True
```

End result flow

1. Drop a pdf in the desired folder (phone / laptop)
2. It gets synced to dropbox and the NAS
3. Hourly the script runs and processes the files
    - Depending on the folder, multiple processors are run
4. When successful, the file is moved to the done folder

Once in a while I check the paperless inbox to process the docs

Take away

- Don't forget to learn while you create/automate
- Having uv makes this a breeze (especially, on a NAS)
- Awesome that something this easy, saves me hours
  - The image snapping bit
  - Dropping attachments from email as well
  - but also finding documents
    - Marking them correctly help enormously

What can you automate in your life?

- Bart Dorlandt
- https://linkedin.com/in/bartdorlandt/
- https://dreamnetworking.nl/
- https://github.com/bartdorlandt/paperless_email_processor
- 14:15 - Camera 5
  - Json freedom or chaos; how to trust your data